

Storm Intensity Distribution (MKM-SI-001)

Wolfram Oracle Benchmark of the Hybrid Normal–Pareto Implementation

MKM Research Labs

Version 0.1 — 20-October-2026

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1 Executive Summary

This document benchmarks the Storm Intensity Distribution (MKM-SI-001): a Normal body joined to a Pareto tail, clamped to $[I_{\min}, I_{\max}]$, with five scenario families from `historical` to `extreme`. It asks whether the code computes what its model document (`storm_intensity.pdf`, v1.3) says, and whether what the document says holds.

The model is not in the production chain. Production storm intensities are drawn in `src/port/src/storm_multi/generators/ batch_generator.py`: an intensity category from `DEFAULT_INTENSITY_WEIGHTS`, then a Normal draw per category from `BASE_INTENSITY_PARAMS`, varied by ±20% within a sequence. Nothing in the port, the routes, the app commands or the scripts imports `IntensityDistribution`; only its own command line and tests do. The model inventory records handovers from SI-001 to SG-001, SS-001, RA-001 and FC-001 that the code does not implement, and its stress scenarios do not reach the storm catalogue, the hazard curves (MKM-GH-001) or PRS pricing (MKM-PR-001).

The code computes its documented formulae. Over 535 cases the model agrees with a 50-digit implementation of the document's equations: exceedance to 8.8×10^{-16}, return-period intensity to 1.3×10^{-16}, and the quantile to 3.8×10^{-9} — the error of Acklam's inverse-normal approximation, carried through $\mu + \sigma z$. Wolfram and the 50-digit reference agree to 4.8×10^{-30}, so the reference is confirmed by a second, independent implementation.

Findings. Of the six claims tested (Table 2), two hold, three do not, and one is a finding:

1. **The clamp is not a slight distortion.** The Pareto tail runs past $I_{\max} = 100$, and every storm it would place there is set to exactly 100: from 1.0% of storms (`historical`) to 29.7% (`extreme`).
2. **So return periods saturate.** The T-year intensity is $I_{\max}$ for every T beyond 4.8 years in the `historical` scenario and 1.13 years in `baseline`; in the stress scenarios, for every return period of interest (Table 4). A 1-in-2-year storm and a 1-in-1000-year storm are the same.
3. **The density is discontinuous at the threshold.** The exceedance is continuous there, as the document proves; the code's docstring further says the Pareto scale makes the *density* continuous, which it does not: the density jumps by a factor of 0.29 to 1.37.
4. **Acklam's accuracy is relative, not absolute.** The document quotes 1.15×10^{-9} absolute error; the measured worst is 8.3×10^{-9} absolute and 1.1×10^{-9} relative, in the far tail. The effect on intensities is negligible.

The sampling jitter changes the mean and standard deviation by at most 0.27%, as the document says.

Evidence. 6 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).

- Code under test, at commit `7bbb63078a59`: `src/models/intensity/distribution/_core.py`, `src/models/intensity/distribution/_math.py`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-20 by `python -m tools.wolfram_oracle si001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`IntensityDistribution` in `src/models/intensity/distribution`: `exceedance_probability`, `inverse_exceedance`, `return_period_intensity` and the Acklam inverse normal behind them, for each of the five `SCENARIO_FAMILIES`, imported as production would import the package.

2.2 What is not

The calibration the document describes (§4.3: maximum likelihood on the body, Hill estimation on the tail) has no code in the repository; the scenario parameters are constants. The inventory's methodology line, "maximum likelihood threshold estimation", contradicts the document's own §8.6 ("no automatic threshold selection"). Neither can be benchmarked. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

With Normal body (μ, σ), threshold u, tail index α and clamp $[I_{\min}, I_{\max}]$, the document defines $p_{\text{tail}} = S_N(u)$, the Normal's exceedance at u, and

$$S(x) = \begin{cases} 1 & x \le I_{\min} \\ S_N(x) & I_{\min} < x \le u \\ p_{\text{tail}}\,(u/x)^{\alpha} & u < x < I_{\max} \\ 0 & x \ge I_{\max} \end{cases} \qquad S^{-1}(p) = \begin{cases} \mu + \sigma\,\Phi^{-1}(1-p) & p \ge p_{\text{tail}} \\ u\,(p_{\text{tail}}/p)^{1/\alpha} & p < p_{\text{tail}} \end{cases} \tag{1}$$

with S^{-1} clamped to $[I_{\min}, I_{\max}]$, and the T-year intensity $S^{-1}(1/(T\lambda))$ for λ storms a year. Above u the tail is a generalised Pareto with shape $1/\alpha$ and scale u/α: a GPD whose scale is tied to the threshold, not fitted.

The clamp. Because the tail is unbounded and the clamp is not a renormalisation, the sampled intensity has point masses $p_{\text{tail}}(u/I_{\max})^{\alpha}$ at $I_{\max}$ and $1 - S_N(I_{\min})$ at $I_{\min}$. The T-year intensity reaches $I_{\max}$ once $1/(T\lambda)$ falls below the first of these, that is for $T > 1/(\lambda\, p_{\text{tail}}(u/I_{\max})^{\alpha})$.

4 Benchmark Method

4.1 References

The equations above are evaluated with `mpmath` at 50 significant digits, from the document rather than the code, with the exact inverse normal. Wolfram evaluates the same expressions from each input's exact value at 60-digit precision. The Normal exceedance is written out as $\frac{1}{2}\operatorname{erfc}\big((x-\mu)/(\sigma\sqrt{2})\big)$: Wolfram 15.0's `SurvivalFunction` of a `NormalDistribution` returns a machine-precision result even for a 60-digit argument, which would have capped the oracle at 16 digits. Acklam's approximation is compared with Wolfram's exact `InverseCDF` on the very doubles it is handed, so input rounding is not counted as approximation error. The moments of the sampled intensity are integrated over p by both, and the jitter's effect is computed in closed form.

4.2 Cases

For every scenario: the exceedance at every 2.5 from $I_{\min}$ to $I_{\max}$; the quantile at $p = 1, 2, 5 \times 10^{-k}$ ($k = 1..6$) and their complements; the return-period intensity for T from 2 to 1000 years. Acklam's approximation alone over both tails down to 10^{-15} and the body in steps of 0.01.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Exceedance $S(x)$ (Eq. 3.4): model vs reference	8.8×10^{-16}	1.0×10^{-12}	pass
Quantile $S^{-1}(p)$, clamped (s3.5): model vs reference	3.8×10^{-9}	1.0×10^{-8}	pass
Return-period intensity (s3.6): model vs reference	1.3×10^{-16}	1.0×10^{-8}	pass
Wolfram agrees with the reference (all cases)	4.8×10^{-30}	1.0×10^{-12}	pass
Acklam Φ^{-1} against Wolfram's exact InverseCDF, absolute	8.3×10^{-9}	see claims	–
Moments $E[Q]$, $E[Q^2]$: Wolfram NIntegrate vs reference	2.0×10^{-16}	1.0×10^{-8}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Document says	Benchmark finds	Outcome
s3.4 S continuous at u	continuous (proved)	jump 1.1×10^{-16}	met
Code docstring: PDF continuous at u	"calibrated so the PDF is continuous"	$f(u^+)/f(u^-)$ from 0.29 to 1.37	**not met**
s3.5, s5.3 Acklam Φ^{-1}	1.15×10^{-9} absolute, full domain	8.3×10^{-9} absolute; 1.1×10^{-9} relative	**not met**
s8.4 Mass put on the clamp	"slight"	up to 29.7% at $I_{\max}$ (extreme)	**not met**
s3.6 Return periods	$S^{-1}(1/T\lambda)$	equal to $I_{\max}$ for every T above 0.17 to 4.84 y, by scenario	finding
s5.4, s8.5 Jitter on the moments	"not material"	0.27% at most	met

5.2 By scenario

Table 3: Each scenario family, exactly: tail probability, the mass the clamp puts on each bound, the return period from which every intensity is $I_{\max}$, the density jump at u, and the moments of the sampled intensity

Scenario	p_{tail}	at $I_{\min}$	at $I_{\max}$	cap from T (y)	$f(u^+)/f(u^-)$	Mean	SD	Jitter on SD
historical	0.048	0.62%	1.0%	4.84	0.29	40.70	13.95	-0.02%
baseline	0.159	0.98%	4.4%	1.13	0.41	47.20	19.26	-0.11%
moderate_stress	0.369	0.38%	11.2%	0.45	0.53	55.35	22.68	-0.18%
severe_stress	0.609	0.62%	21.5%	0.23	0.86	62.93	25.90	-0.23%
extreme	0.773	0.62%	29.7%	0.17	1.37	68.10	26.53	-0.27%

Table 4: Return-period intensity, $\lambda = 20$ storms a year (reference; the model agrees to the figures in Table 1). "cap" means $I_{\max}$

Scenario	$T = 2$	$T = 5$	$T = 10$	$T = 25$	$T = 50$	$T = 100$	$T = 200$	$T = 500$	$T = 1000$
historical	74.46	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**
baseline	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**
moderate_stress	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**
severe_stress	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**
extreme	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**	**cap**

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute the documented formulae; whether a Normal–Pareto hybrid suits storm intensity, and whether the scenario parameters are calibrated, are separate questions.

2. **Wiring.** Whether SI-001 should drive the storm catalogue, or the inventory's handovers should be corrected to the code, is for the model owner and the MRC.

3. **The clamp.** Renormalising the tail on $[u, I_{\max}]$, or raising $I_{\max}$, would remove the point mass and restore informative return periods; either changes the model and is for the owner.

4. **Documentation.** §3.5 and §5.3 should quote Acklam's accuracy as relative; §8.4 should state the masses above; the code's docstring should drop the claim of a continuous density.

7 Document History

Date	Version	Description	Author
20-Oct-2026	0.1	First issue	David K Kelly